The Evolution of Home Insulation: A Century of Advancements

Malik Husein[1], Andrew Earls[2], Seif Al Din Husein[3], Lance Lease[4]

[1]m.m.husein@vikes.csuohio.edu, [2]a.earls@vikes.csuohio.edu, [3]s.m.husein14@vikes.csuohio.edu, [4]l.lease@vikes.csuohio.edu

Choose Ohio First

Abstract:

This project examines the evolution of home insulation over the past century, with a focus on the development, effectiveness, and impact of various insulation materials. While basic insulation techniques date back thousands of years, modern insulation as we know it emerged in the 1930s and became widely adopted by the 1950s. Since then, materials such as fiberglass, cellulose, mineral wool, foam, and aerogel have been introduced to improve thermal resistance. The primary function of insulation is to regulate indoor temperatures by resisting heat transfer. This resistance is quantified by the R-value, which indicates how well a material slows the flow of heat — the higher the R-value, the better the insulating performance. This study explores how advances in insulation technology have enhanced energy efficiency and thermal performance, highlighting historical trends, breakthroughs in material science, and their influence on contemporary home construction.

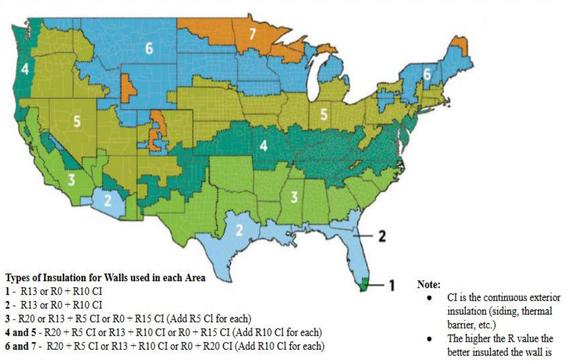
Objectives:

- Analyze the evolution of insulation materials
- Assess the impact of R-value changes on thermal performance
- Evaluate insulation effectiveness across various climate zones
- Explore the health and environmental

Methodology:

- Observational Analysis: Examining historical and current insulation practices and installations
- Case Studies: Reviewing specific homes or buildings representing different eras and insulation types
- Content Analysis: Analyzing historical documents, product manuals, and marketing materials related to insulation
- Secondary Data Analysis: Interpreting existing data from energy efficiency reports, R-value studies, and environmental assessments

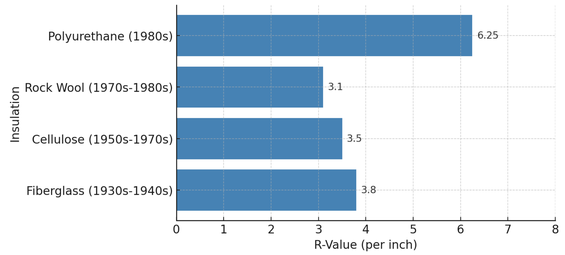
Climate Conditions:



In colder climates, higher R-values are necessary to retain heat and improve energy efficiency, while in warmer regions, lower R-values may be sufficient to reduce heat gain and maintain indoor comfort.*

*Department of Energy. (2022). Insulation. Energy.gov., https://www.energy.gov/energysaver/insulation

Historical R-Values of Insulation



Student Guide. (2017). Need.org., https://www.need.org/Files/chemistry/ChemistryStudentGuideInsulation.html

Fiberglass:
- Composed of plastic reinforced with fine glass fibers [5]
- Cost-effective and widely used [1]
- Allows airflow, which can reduce energy efficiency and lead to uncomfortable indoor conditions. [1]

Cellulose:
- Made from recycled paper products (high recycled material content) [5]
- Dense structure generally provides good thermal resistance [2]
- Can settle or compress over time, potentially reducing its R-value [2]

Rock wool(Mineral Wool):
- Consists of fine, interwoven fibers bound with starch, available as loose fill or batts [3]
- Provides good fire resistance and sound insulation [3]
- Retains moisture that may affect performance and indoor air quality [3]

Polyurethane:
- A polymer formed by reacting organic cyanates with polyols and additives [4]
- Lightweight with low thermal conductivity and high insulating performance [4]
- Increasingly used across various industries, including construction, aerospace, and biomedicine [4]

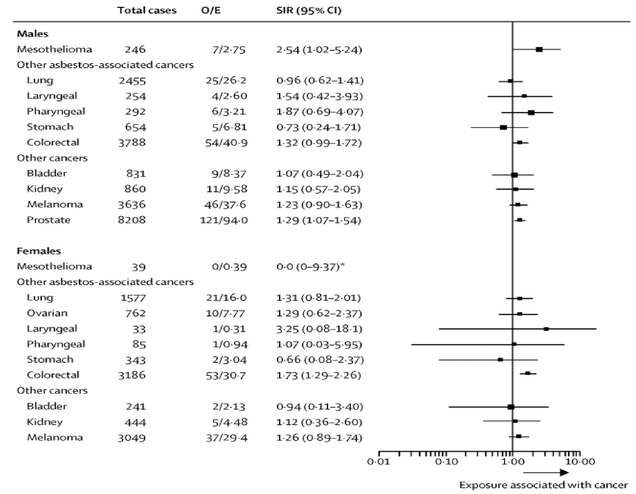
[1] RetroFoam of Michigan Inc. (2017). What is Fiberglass Insulation? How it Works and What it's Made of Retrofoamofmichigan.com. https://www.retrofoamofmichigan.com/blog/fiberglass-insulation-material-ingredients
[2] *webgrowth. (2022, July 11). What Is Cellulose Insulation? Pros and Cons. Attic Construction. https://atticconstruction.com/what-is-cellulose-insulation-pros-and-cons/*
[3]*Rockwool for Insulation. (n.d.). Www.nachi.org., https://www.nachi.org/rockwool.htm*
[4] *Wang, Z., Wang, C., Gao, Y., Li, Z., Shang, Y., & Li, H. (2023). Porous Thermal Insulation Polyurethane Foam Materials. Polymers, 15(18), 3818. https://doi.org/10.3390/polym15183818*
[5] Insulation Materials. (2025). Energy.gov., https://www.energy.gov/energysaver/insulationmaterials#fiberglass%E2%80%8B

Future Work:

The future of insulation is being driven by the development of innovative materials that emphasize both sustainability and performance. Bio-based insulation—derived from recycled textiles, agricultural waste, and other renewable sources—is gaining popularity for its eco-friendliness and effective thermal properties.* Mycelium-based materials, cultivated from fungal networks, are emerging as biodegradable and thermally efficient alternatives.* Additionally, ongoing advancements in aerogels—known for their ultra-lightweight structure and exceptional insulating capabilities—are setting new standards for thermal management in buildings.* These trends reflect a growing shift in the building industry toward greener, more efficient insulation solutions.

*Petcu, C., Hegyi, A., Stoian, V., Dragomir, C. S., Ciobanu, A. A., Lăzărescu, A.-V., & Florean, C. (2023, August 3). *Research on thermal insulation performance and impact on indoor air quality of cellulose-based thermal insulation materials*. Materials (Basel, Switzerland).

Cancer Risk by Asbestos Exposure



Norman, G. R., & Newhouse, M. T. (1986). Health effects of urea formaldehyde foam insulation: evidence of causation. PubMed, 134(7), 733–738.

Several insulation materials can pose health risks to those who come into contact with them. Asbestos-based insulation, in particular, is a known carcinogen that was widely used in the U.S. during the mid-20th century.* The accompanying chart highlights the increased incidence of cancer linked to asbestos exposure.

*Korda, R. J., Clements, M. S., Armstrong, B. K., Law, H. D., Guiver, T., Anderson, P. R., Trevenar, S. M., & Kirk, M. D. (2017). Risk of cancer associated with residential exposure to asbestos insulation: a whole-population cohort study. The Lancet Public Health, 2(11), e522–e528. https://doi.org/10.1016/s2468-2667(17)30192-5

Conclusion:

The evolution of home insulation over the past century has led to substantial improvements in thermal efficiency, safety, and overall building performance. Early materials like asbestos and rock wool provided basic insulation but came with serious health concerns. Breakthroughs such as fiberglass in the 1930s, cellulose in the 1950s, and polyurethane spray foam in the 1980s marked significant progress by improving R-values and ease of application. Today, insulation materials continue to evolve—striking a balance between thermal performance, environmental impact, and occupant health. Looking ahead, future innovations are expected to further enhance sustainability, durability, and energy efficiency in residential construction.

Acknowledgments:

We sincerely thank Dr. Navid Goudarzi, Principal Investigator of the Multiscale Multiphysics Modeling and Data-Driven Analysis of Thermofluids (M^3TFluiD) Lab in the Department of Mechanical Engineering at CSU, for his invaluable guidance and support throughout our research project.